Sources

IAEA LiveChart of Nuclides
ENSDF evaluations; export of 18 October 2023

NUBASE2020
Kondev et al. 2021; 2,019 metastable isomers

Decay-mode definitions
ΔZ and ΔA for 32 modes (NUBASE2020 notation)

ICRP Publication 107
Nuclear decay data for dosimetric calculations (2008)

IUPAC · OCE · SNOMED CT
DICOM PS3.16: CID 18, CID 4020 (release 2026c)

Build (scripted, reproducible)

1 Load ground states
Half-life in seconds (ħ ln2 / Γ for resonances); spin-parity, abundance, decay modes.

2 Generate isomers
NUBASE state index → Tc-99m, Lu-177m3 …; excitation energy, half-life, decay modes.

3 Create decay branches, derive daughters
One :DecayBranch per decay mode, with its ratio.
Z′ = Z + ΔZ, A′ = A + ΔA → daughter ground state
No daughter for fission; cluster decay subtracts the cluster. 7,399 of 7,691 branches linked.

4 Add emissions
Type, energy and yield per decay (≥ 0.1%, ≥ 1 keV) for DICOM-coded nuclides and their decay chains.

5 Map external codes
skos:exactMatch to OCE and SNOMED CT; :dicomCode as in the Radionuclide Code Sequence.

6 Mirror values onto classes
Display copies on each punned class (:displayValue) for viewers such as BioPortal.

Validation

Internal consistency
SPARQL checks V1–V6: nucleon conservation, A = Z + N, no self-decay, no dangling links, element Z, display copies match. HermiT: consistent, no unsatisfiable classes.

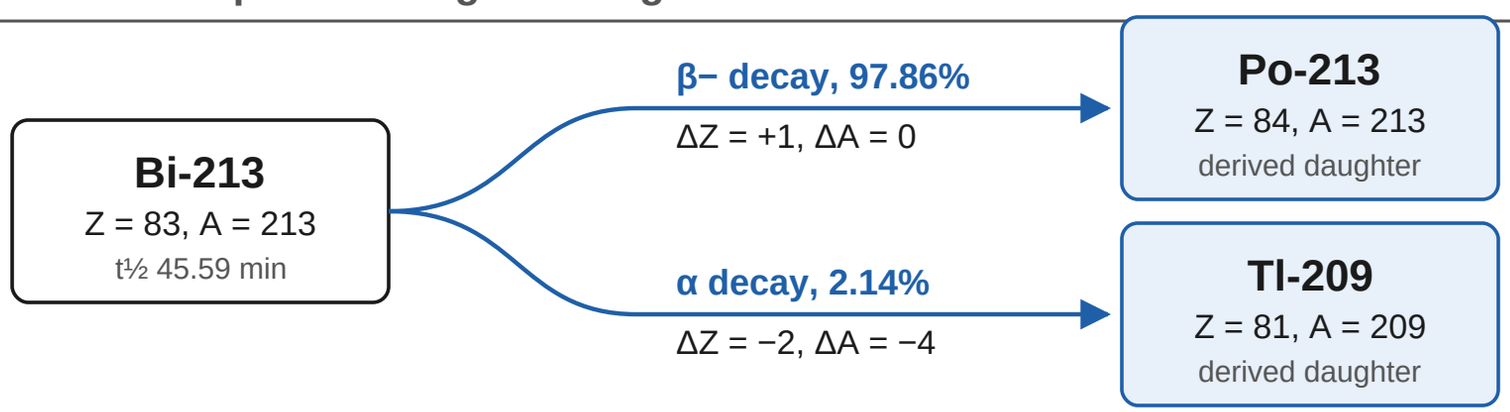
Daughters vs independent data
ICRP-107: 1,510 links agree
ENSDF: 1,688 links agree
0 disagreements

Values vs sources
IAEA export: 6 documented exceptions
Gamma lines vs ENSDF: 204 of 206 agree

Manual audit
Stratified random sample of 100 nuclides, two independent reviewers

Release: NUCLEON.ttl (OWL 2 DL)
118 elements · 5,408 nuclides (2,019 isomers)
7,691 decay branches · 3,990 emissions
BioPortal · GitHub: scripts, queries, reports

Worked example: deriving the daughters of Bi-213



Each daughter is computed from the parent's Z and A and the mode's ΔZ and ΔA, never transcribed, so every link can be re-derived and checked (query V1).